ZigBee
The open source ZigBee protocol is widely touted to be the standard for home automation applications. The protocol allows third parties to offer services based on inputs from appliances at home

and set for the curtains to be opened, and save this as your morning setting, and of course have it come on at a particular time, just as your alarm goes off. Similarly, you could save different settings for "watching a movie", and so on.

The control extends out to the general building, in that the whole site is managed by an Intelligent Building Management System. This is the basic control system which takes care of power, fire and water management. A key factor in the overall management is of course, security. For the overall system, one interesting feature is the ability to check during the night if the guards are awake! But it is the security of the individual homes that is most interesting.

Touch screen access to the main control system. Top left window shows view from the front entrance.

The first security feature that you notice is the front door control. The lock itself can be controlled in three ways – by a magnetic card that you hold in front of a reader, by the main control system in the house – which you could, if you really wanted to, access from a mobile phone as you approach the door – and by a good old-fashioned key. The key is the most likely way one would secure the apartment when it is empty, as the electronic control only engages a latchbolt, whereas the key engages first the latchbolt and then three deadbolts.

It seems that the electronic control would be best used when the home is occupied. When somebody comes to the door and rings the bell, you see them from the camera at the front door and can speak to them and then release the door catch. Usefully, their image is captured. This also happens when the home is unoccupied, and when you return home you have a complete set of images of all the people who have come near the front door, even if they have not rung the bell.

As the system can be accessed remotely, you can call it up across the internet, and access, for example, the images captured at the front door since you left. This may not seem immediately useful, but security devices are for unexpected circumstances. Similarly, you can have cameras installed inside the home, and these can also be accessed remotely. You can call up your system, turn on the lights and inspect the rooms. When might you do this? Just to give yourself peace of mind, perhaps; to check that the kids have not wrecked the place with a party, and also when some alarm has gone off.

There are sensors in the home that detect motion inside and if a window has been broken. These intrusion alarms can trigger SMSs or emails, and you can then call into the home and see what has been going on. Similarly, there is a gas leak detector that will also send a message so that you can then call up the home, turn off the gas supply, and then turn on the cameras to check if there has been a fire. The detector should capture the presence of gas before enough accumulates that turning anything on or off might cause ignition!

In the second eHome that I visited, I noticed no internal cameras. Surprisingly, these are now optional extras, as many customers have complained that they are an intrusion of privacy, even though they are under the complete control of the occupants.

Conclusions

We looked at these eHome projects from Designarch as an example of the direction in which electronic control and security are moving. The individual products used are not the key features here – there are many suppliers that can equip a home with controllable cameras, switches for internal systems such as heaters and curtains, and so on. The main control system featured in the eHomes is by Homesdigital (**www.zeos.co.uk**), a company that is based mainly in the UK and India.

Increasingly, the prices for all of these products are falling and there is no doubt that demand is rising. So, you could assemble most of what has been described here and have it installed in your own home. However, to see it all in one place, then the first projects in India of which we are aware are these ones from Designarch. No doubt many others will follow in the near future.

In my view the most impressive aspect of the eHomes projects is certainly the security. This has been well thought out and seems comprehensive. I asked about the reliability of the systems, and was told that where possible, multiple means of control are in place. The most obvious was the front door lock. Not only do you have both electronic and key access, but there are two electronic systems, and they are separate. So, the magnetic card works in parallel to, and not through, the central control system, and if the latter should have failed – it is a PC running Windows, after all – the card should still work, and vice-versa.

The type of setup here would be suitable for people wanting a modern electronic home, but not with particularly demanding comuting needs or expectations. More demanding home owners would no doubt expect Ethernet to come as standard – the most surprising thing missing from these particular projects.


Kitchen and dining area. Not visible in this photo, a TV image can be displayed on the mirror.